

Wayfair E-commerce Sales & Profitability Analysis

1. Project Overview

Wayfair operates a large e-commerce marketplace where sales performance can vary across products, categories and time periods. This analysis evaluates historical sales data from 2024-2025 to understand changes in sales and profitability, identify products and categories contributing to performance and examine the roles of returns and cancellations.

This project uses a simulated dataset based on a Wayfair business scenario and is intended to demonstrate an end-to-end data analytics workflow.

Business Questions:

1. How has sales performance changed from 2024-2025?
2. Which products and categories drive revenue and profitability?
3. How are returns and cancellations affecting performance?

2. Dataset & Methodology

2.1 Dataset Overview

This project uses four datasets covering customers, products, orders and order items. The data provides information needed to analyse sales, profitability, product performance and order status from 2024-2025.

Dataset	Records	Grain	Purpose
Customers	3,600	One row per customer	Customer and acquisition information
Products	90	One row per product	Product, category, price and cost information
Orders	9,207	One row per order	Order dates, status, payment, discounts and shipping
Order items	15,409	One row per product line within an order	Product-level transaction details

2.2 Tools & Technologies

The analysis was completed using:

- **Python** – Data Understanding, Cleaning and Exploratory Analysis
- **PostgreSQL** – Data Modelling, KPI calculations and Business-question Analysis
- **Power BI** – Interactive Dashboard and Visualization

2.3 Analytical Approach

The project followed an end-to-end analytical workflow:

Raw Data → Data Cleaning → Exploratory Data Analysis → Data Modelling → KPI Definition → SQL Analysis → Power BI Dashboard → Insights & Recommendations

The analysis focused on three key areas: sales performance over time, product and category profitability and the impact of returns and cancellations.

3. Data Cleaning & Data Modelling

3.1 Data Cleaning

The raw datasets were reviewed for missing values, duplicate records, invalid values and data-type issues before analysis.

The main cleaning steps included:

- Removed **38 exact duplicate order records**.
- Removed **199 order-item records** with missing unit prices.
- Converted order and customer data fields from text to date format.
- Replaced missing discount codes with **"No Discount"**.
- Standardized product category names by removing extra spaces and applying consistent capitalization.
- Validated relationships between customers, orders, order items and products.

After cleaning, the final datasets contained **3,600 customers, 9,207 orders, 15,409 order items and 90 products**.

3.2 Derived Financial Metrics

To support sales and profitability analysis, three financial metrics were derived from the transaction-level data:

Revenue = (Quantity x Unit Price) – Discount Amount

Cost = Quantity x Unit Cost

Profit = Revenue – Cost

These derived metrics were stored in the fact_order_items table and used consistently across the SQL analysis and Power BI dashboard.

3.3 Data Modelling

A relational data model was created in PostgreSQL to support consistent analysis and reporting. The model consists of customer, product, date and order dimensions connected to the order-item fact table.

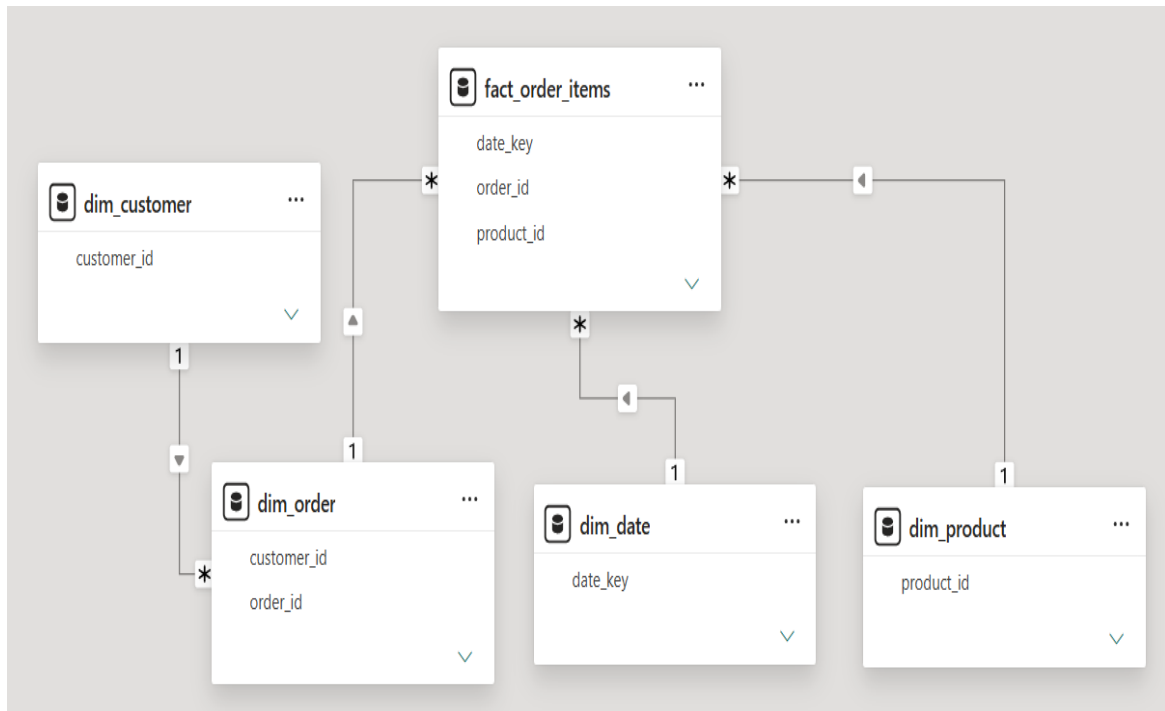


Fig 1 – Power BI data model for the Wayfair E-commerce Sales & Profitability Analysis

The fact table contains transaction-level quantities, prices, discounts, revenue, cost and profit. Dimension tables provide descriptive information used to analyse these transactions by customer, product, date and order status.

4. Analysis & Key Findings

The business questions were analysed using PostgreSQL against the dimensional data model. The analysis covers sales performance over time, category and product profitability and return and cancellation behaviour.

4.1 Sales Performance Over Time

Year-over-Year Sales & Profitability

	year integer	total_revenue numeric	total_profit numeric	profit_margin numeric
1	2024	926503.78	442398.70	47.75
2	2025	1120810.21	564779.79	50.39

Revenue increased from approximately \$926.5K in 2024 to \$1.12M in 2025, while profit increased by 27.7%. Profit margin also improved from 47.75% to 50.39%.

Key Finding:

2025 showed stronger sales and profitability, with profit growing faster than revenue and overall profit margin improving.

Monthly Revenue & Profit Trend

	year integer	month integer	month_name character varying (20)	total_revenue numeric	total_profit numeric
1	2024	1	January	61835.60	30371.06
2	2024	2	February	66811.35	32326.22
3	2024	3	March	77617.06	37216.83
4	2024	4	April	69247.95	33708.70
5	2024	5	May	81418.16	38838.48
6	2024	6	June	72056.04	35320.28
7	2024	7	July	84294.91	40972.66
8	2024	8	August	67663.46	32823.83
9	2024	9	September	71244.45	34714.16
10	2024	10	October	69844.83	34330.65
11	2024	11	November	107507.48	48334.02
12	2024	12	December	96962.49	43441.81
13	2025	1	January	78143.35	40057.76
14	2025	2	February	75430.60	38428.09
15	2025	3	March	85946.54	43315.85
16	2025	4	April	86177.99	43364.28
17	2025	5	May	90989.39	46392.36
18	2025	6	June	100794.63	51990.49
19	2025	7	July	91074.58	46972.10
20	2025	8	August	99330.73	50921.23
21	2025	9	September	82579.54	42480.23
22	2025	10	October	90329.70	46604.41
23	2025	11	November	141099.04	66733.62
24	2025	12	December	98914.12	47519.37

Monthly revenue and profit were analysed across 2024 and 2025 to identify changes in sales performance throughout the year. November recorded the strongest revenue performance in both years.

Key Finding:
Sales performance varies throughout the year, with November representing a significant seasonal peak.

Year-over-Year Monthly Revenue Comparison

	month integer	month_name text	revenue_2024 numeric	revenue_2025 numeric	revenue_growth_percent numeric
1	1	January	61835.60	78143.35	26.37
2	2	February	66811.35	75430.60	12.90
3	3	March	77617.06	85946.54	10.73
4	4	April	69247.95	86177.99	24.45
5	5	May	81418.16	90989.39	11.76
6	6	June	72056.04	100794.63	39.88
7	7	July	84294.91	91074.58	8.04
8	8	August	67663.46	99330.73	46.80
9	9	September	71244.45	82579.54	15.91
10	10	October	69844.83	90329.70	29.33
11	11	November	107507.48	141099.04	31.25
12	12	December	96962.49	98914.12	2.01

Monthly revenue was compared between 2024 and 2025 to identify which months experienced increased or decreased in sales. November revenue increased from approximately \$107.5K in 2024 to \$141.1K in 2025.

Key Finding:
The month-level comparison highlights year-over-year changes and provides evidence of seasonal revenue patterns.

4.2 Product & Category Performance

Category Revenue & Profitability

	category character varying (100) 🔒	total_revenue numeric 🔒	total_profit numeric 🔒	profit_margin numeric 🔒
1	Furniture	367852.11	170306.95	46.30
2	Kitchen	301330.45	142521.19	47.30
3	Decor	300056.43	153729.50	51.23
4	Bedding	289352.35	148239.17	51.23
5	Outdoor	270559.53	135052.49	49.92
6	Lighting	251863.08	119535.43	47.46
7	Storage	148643.90	82636.81	55.59
8	Bath	117656.14	55156.95	46.88

Furniture generated the highest revenue at approximately \$367.9K and the highest profit at approximately \$170.3K. Storage achieved the highest profit margin at 55.59%, compared with Furniture's 46.30%.

Key Finding:

The highest-revenue category is not necessarily the highest-margin category, so both sales volume and profitability should be considered when evaluating category performance.

Top 10 Products by Revenue

	product_id integer 🔒	product_name character varying (255) 🔒	category character varying (100) 🔒	total_revenue numeric 🔒	total_profit numeric 🔒	profit_margin numeric 🔒
1	1059	Cotton Patio Chair	Outdoor	74461.90	33252.34	44.66
2	1042	Classic Dining Chair Pair	Furniture	59719.14	25417.95	42.56
3	1072	Coastal Picture Frame Set	Decor	47657.93	26689.41	56.00
4	1041	Nordic Side Table	Furniture	45961.79	25540.47	55.57
5	1040	Marble Bookshelf	Furniture	45225.82	19615.52	43.37
6	1044	Bamboo Nightstand	Furniture	44543.02	18783.97	42.17
7	1018	Ceramic Quilt	Bedding	44446.23	22793.43	51.28
8	1076	Modern Wreath	Decor	43522.33	22272.43	51.17
9	1080	Classic Garland	Decor	42899.86	24144.66	56.28
10	1054	Velvet String Lights	Lighting	40448.35	22827.60	56.44

The analysis ranks the top 10 products based on total revenue. Cotton Patio Chair generated approximately \$74.5K in revenue and \$33.3K in profit, making it the highest-revenue product.

Key Finding:

A small group of products contributes significantly to sales, making the top-revenue products important contributors to overall business performance.

Top 10 Products by Profit

	product_id integer	product_name character varying (255)	category character varying (100)	total_revenue numeric	total_profit numeric	profit_margin numeric
1	1059	Cotton Patio Chair	Outdoor	74461.90	33252.34	44.66
2	1072	Coastal Picture Frame Set	Decor	47657.93	26689.41	56.00
3	1041	Nordic Side Table	Furniture	45961.79	25540.47	55.57
4	1042	Classic Dining Chair Pair	Furniture	59719.14	25417.95	42.56
5	1080	Classic Garland	Decor	42899.86	24144.66	56.28
6	1054	Velvet String Lights	Lighting	40448.35	22827.60	56.44
7	1018	Ceramic Quilt	Bedding	44446.23	22793.43	51.28
8	1076	Modern Wreath	Decor	43522.33	22272.43	51.17
9	1063	Matte Black Hammock	Outdoor	40434.01	22225.28	54.97
10	1052	Linen Pendant Light	Lighting	36777.02	21178.14	57.59

Products were separately ranked by total profit to identify the products making the largest contribution to profitability. This provides a different perspective from the revenue ranking.

Key Finding:

Comparing revenue leaders with profit leaders helps distinguish products that generate high sales from those that generate strong financial returns.

4.3 Returns & Cancellations

Revenue & Profit by Order Status

	status character varying (30)	total_orders bigint	total_revenue numeric	total_profit numeric
1	completed	8397	1858121.00	914173.10
2	returned	403	114722.42	56764.51
3	canceled	340	74470.57	36240.88

Revenue, profit and order counts were analysed across order statuses to understand how different order outcomes contribute to overall business performance.

Key Finding:

Order status provides additional context for interpreting revenue and profit, particularly when evaluating completed, returned and cancelled orders.

Return & Cancellation Rate by Category

	category character varying (100)	total_orders bigint	returned_orders bigint	canceled_orders bigint	return_rate numeric	cancellation_rate numeric
1	Furniture	995	116	31	11.66	3.12
2	Bedding	1779	89	58	5.00	3.26
3	Bath	1624	71	73	4.37	4.50
4	Outdoor	1333	58	50	4.35	3.75
5	Kitchen	2812	118	107	4.20	3.81
6	Decor	2696	112	93	4.15	3.45
7	Storage	1710	70	56	4.09	3.27
8	Lighting	1211	48	36	3.96	2.97

Return and cancellation rates were calculated for each product category to identify categories with higher levels of returned and cancelled orders.

Key Finding:

Furniture recorded the highest return rate at 11.66% while Bath recorded the highest cancellation rate at approximately 4.50%.

4.4 Power BI Dashboard

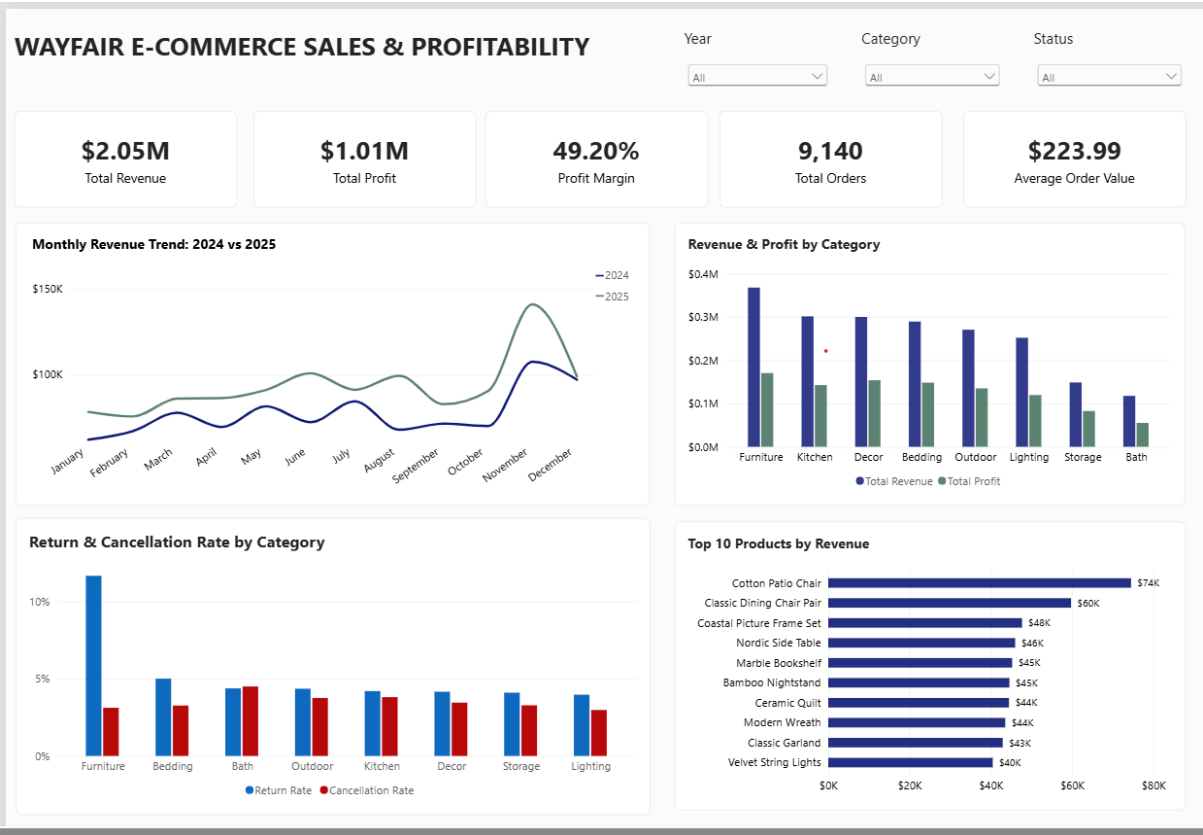


Fig 2 – Power BI Dashboard summarizing sales performance, profitability, product performance and order-status analysis

The Power BI dashboard brings together the key findings from the analysis through interactive KPIs, trend analysis, category and product performance and return and cancellations rates. Users can filter the analysis by year, category and order status to explore different aspects of business performance.

5. Business Recommendations

- **Monitor category-level profitability:** Track revenue together with profit margin to identify categories where strong sales volume does not necessarily translate into higher profitability.
- **Investigate Furniture returns:** Furniture has the highest return rate at **11.66%**. Further analysis should examine potential causes such as product expectations, product descriptions, quality or fulfilment.
- **Review Bath cancellations:** Bath has the highest cancellation rate at approximately **4.50%**. Investigating the reasons for cancellations could help identify operational or product-related issues.
- **Evaluate high-revenue products:** Products generating high revenue but comparatively lower profit margins should be reviewed for pricing, cost and promotional strategies.
- **Plan around seasonal demand:** November recorded the highest revenue and order volume in both years. Seasonal demand patterns should be considered when planning inventory and promotional activity.

6. Limitations

- The dataset represents a simulated e-commerce business scenario and does not represent internal Wayfair data.
- 199 order-item records with missing unit prices were removed, which means some transactions are not included in transactional-level revenue and profitability analysis.
- The dataset does not contain refund amounts or the financial impact of cancellations and returns, so their direct monetary impact could not be calculated.
- The analysis covers only the 2024-2025 period, so longer-term trends could not be evaluated.

7. Conclusion

This analysis evaluated e-commerce sales performance, profitability, product and category performance and order-status patterns for 2024-2025. The result showed overall improvement in revenue and profitability in 2025, with Furniture contributing the highest revenue and profit. The analysis also identified differences in return and cancellation rates across categories, highlighting areas for further investigation.

The Power BI dashboard brings these findings together into an interactive view that can support ongoing monitoring of sales, profitability and order performance.